

Jing Peng

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Education

Xi'an Jiaotong University

Bachelor of Engineering

Qian Xuesen Honors Class, major in Automation and minor in Electrical Engineering

○ Rank – **1st**; GPA – **4.0**

Xi'an, China

September 2021 – June 2025

University of California, Berkeley

Exchange Student

Berkeley Global Access Program (BGA), College of Engineering

○ Courses – CS61B(A), Mec136(A), EECS149, ELENG113(A-)

Berkeley, USA

August 2023 – December 2023

Shanghai Jiao Tong University

Ph.D. Student

X-LANCE Lab, College of Computer Science, Zhiyuan Honor Ph.D. Program

○ Advisors – Prof. Kai Yu and Prof. Shinji Watanabe (Carnegie Mellon University, USA); close collaboration with Prof. Shuai Wang

Shanghai, China

September 2025 – Present

Research Interests

○ Speech Large Language Models (Speech LLMs): speech-text alignment, low-resource adaptation, multilingual and multi-speaker understanding, audio agents, and reproducible evaluation; streaming ASR, error correction, and reliability.

Research Experience

Reward Model Design and Optimization in RLHF

Feb 2024 – May 2024

- Designed reward models for alignment in Reinforcement Learning from Human Feedback
- Research conducted at PKU Alignment Lab, Peking University

Research on Speech Large Language Models

Jul 2024 – Present

- Developed speech-understanding methods that connect speech representations with language-model reasoning and instruction following
- Internship project at AISpeech, Suzhou, Jiangsu

Speech Large Language Models for SA-ASR

Sep 2025 – Present

- Research intern on Speaker-Attributed ASR with Speech LLMs, improving speaker consistency and controllability in multi-speaker recognition
- Internship project at Shenzhen Research Institute of Big Data, Remote

Speaker Discrimination on Omni and SLM

Sep 2025 – Present

- Focused on explicit speaker discrimination, implicit speaker selection, and speaker-identity modeling for omni-modal and speech-language models
- Internship conducted at Hi Lab, Xiaohongshu, Shanghai

Open Source Projects

Open-Bench [Website] & SURE [Demo]

Lead

- Led the unified experimentation platform and its interactive evaluation demo.

- sure-eval** [[GitHub](#)] & **sure-evaluation** [[GitHub](#)] Lead
- Led development of the open-source evaluation framework and integrated evaluation pipeline.
- paper_claw** [[GitHub](#)] Lead
- Developed paper-discovery utilities with personalized research digests, classification, and multilingual summaries.
- ps-slm** / **TASU** [[GitHub](#)] Lead
- Led the open-source implementation of text-only Speech LLM alignment for ASR and speech understanding.
- Qwen-Audio-Agent** [[GitHub](#)] Core Contributor
- Contributed architecture optimization and backend model expansion for the audio agent framework.
- SLAM-LLM** [[GitHub](#)] Contributor
- Contributed NPU support to the unified speech and language model training framework.
- Hugging Face speech-to-speech** [[GitHub](#)] Contributor
- Contributed CJK punctuation preservation and codec-token budgeting for Chinese text-to-speech.

Selected Publications

* indicates equal contribution.

Published and Accepted Papers

- **Jing Peng***, C. Wang*, Y. Yang, L. Qian, J. Li, Y. Xi, S. Wang, K. Yu. *TASU2: Controllable CTC Simulation for Alignment and Low-Resource Adaptation of Speech LLMs*. arXiv:2604.08384, 2026. (Accepted by INTERSPEECH 2026, Oral)
- **Jing Peng***, J. Du*, C. Wang*, H. Li*, Y. Yang*, et al. *A Unified and Reproducible Experimentation Framework for Speech Understanding (SURE)*. arXiv:2605.30899, 2026. (Accepted by INTERSPEECH 2026, Oral)
- W. Huang, Y. Qiu, B. Li, Y. Guo, **Jing Peng**, H. Wang, X. Chen, K. Yu. *RAS: A Reliability Oriented Metric for Automatic Speech Recognition*. arXiv:2604.24278, 2026. (Accepted by INTERSPEECH 2026, Oral)
- W. Tu, J. Gao, Y. Huo, Y. Wang, **Jing Peng**, B. Li, Z. Ma, T. Liu, S. Fan, K. Yu, X. Chen, Z. Zheng. *VISA: A Visual Information Strengthened Audio-Reasoning System for the Interspeech 2026 ARC Agent Track*. arXiv:2606.07264, 2026. (Accepted by INTERSPEECH 2026, ARC Agent Track, Oral)
- **Jing Peng***, Z. Chen*, H. Li*, Y. Wang, D. Ma, M. Li, Y. Du, D. Xu, K. Yu, S. Wang. *G-STAR: End-to-End Global Speaker-Tracking Attributed Recognition*. arXiv:2603.10468, 2026. (Accepted to Findings of EMNLP 2026)
- **Jing Peng***, Q. She*, Y. Fang, Y. Xi, K. Yu. *TC-BiMamba: Trans-Chunk Bidirectionally within BiMamba for Unified Streaming and Non-Streaming ASR*. arXiv:2602.11546, 2026. (Accepted to Findings of EMNLP 2026)
- **Jing Peng***, Y. Wang*, Y. Fang, Y. Xi, X. Li, X. Zhang, K. Yu. *A Survey on Speech Large Language Models for Understanding*. arXiv:2410.18908, 2024. (Published in IEEE JSTSP, 2025)
- **Jing Peng**, Y. Yang, X. Li, Y. Xi, Q. Tang, Y. Fang, J. Li, K. Yu. *TASU: Text-Only Alignment for Speech Understanding*. arXiv:2511.03310, 2025. (Accepted by ICASSP 2026, Oral)
- Y. Fang*, **Jing Peng***, X. Li, Y. Xi, C. Zhang, G. Zhong, K. Yu. *Low-Resource Domain Adaptation for Speech LLMs via Text-Only Fine-Tuning*. arXiv:2506.05671, 2025. (Accepted by ASRU 2025)
- Junjie Li, **Jing Peng**, Yangui Fang, Shuai Wang, Kai Yu. *MOSA: Mixtures of Simple Adapters Outperform Monolithic Approaches in LLM-based Multilingual ASR*. arXiv:2508.18998, 2025. (Accepted by ICASSP 2026, Poster)

- Y. Fang, B. Cheng, **Jing Peng**, X. Li, Y. Xi, C. Zhang, G. Zhong. *Fewer Hallucinations, More Verification: A Three-Stage LLM-Based Framework for ASR Error Correction*. arXiv:2505.24347, 2025. (Accepted by ASRU 2025)
- B. Li, W. Huang, Y. Qiu, Y. Guo, H. Wang, Z. Li, **Jing Peng**, Z. Ma, X. Chen, K. Yu. *ISA-Bench: Benchmarking Instruction Sensitivity for Large Audio Language Models*. arXiv:2510.23558, 2025. (Accepted by ICASSP 2026, Oral)

Additional Accepted Papers: NCMMSC 2026

Author lists and public links are TBD.

- SURE-EVAL: *A Systematic and Unified Agentic Framework for Reproducible Evaluation*. (Accepted by NCMMSC 2026, Oral)
- Multilingual TASU: *Extending Text-only Alignment to Multilingual Speech Recognition via Simulated CTC Posteriors*. (Accepted by NCMMSC 2026, Oral)
- MoE Architectures for Transformer Encoder-Based EEND: *Towards Scalable Multi-Speaker Diarization*. (Accepted by NCMMSC 2026, Oral)
- PA-TSE: *Presence-Aware Target Speaker Extraction for Complex Sparse-Overlap Conversations*. (Accepted by NCMMSC 2026, Oral)
- SCORE: *Semantic Consequence-Oriented Recognition Evaluation for Context-Aware ASR*. (Accepted by NCMMSC 2026, Poster)
- DiS-TSE: *Spatial-Cue-Preserving Multichannel-to-Multichannel Target Speech Extraction with Direction Clues*. (Accepted by NCMMSC 2026, Poster)
- Agentic Data Sampling Framework for Multilingual ASR via Linguistic-Aware Error-Driven Optimization. (Accepted by NCMMSC 2026, Poster)

Preprints and Submissions

- Y. Wang*, **Jing Peng***, H. Li, C. Wang, W. Tu, Y. Xi, Z. Sun, K. Yu, S. Wang. *Audio-Mind: An Auditable Agentic Framework for Audio Understanding*. arXiv:2605.28480, 2026. (Submitted to EMNLP 2026)
- H. Li*, **Jing Peng***, Z. Wang, L. Chen, K. Yu. *XFlow: An Executable Protocol Programming System for Reliable Multi-Agent Workflows*. arXiv:2606.14790, 2026.
- Z. Wang, H. Li, Z. Yang, Z. Hu, S. Zuo, Y. Zhang, D. Ma, D. Luo, C. Wang, **Jing Peng**, T. Huang, S. Guo, H. Wang, Z. Zhu, S. Han, Y. Cao, K. Yu, L. Chen. *Externalizing Research Synthesis and Validation in AI Scientists through a Research Harness*. arXiv:2606.18874, 2026.
- Y. Fang*, **Jing Peng***, Y. Xi, X. Li, H. Li, C. Zhang, G. Zhong, K. Yu. *Joint Decoding Method for Controllable Contextual Speech Recognition Based on Speech LLM*. arXiv:2508.08585, 2025.

Academic Honors

Merit Student, Xi'an Jiaotong University	2022 – 2024
○ Recognized for academic excellence, campus engagement, and leadership	
Outstanding Graduate Student, Xi'an Jiaotong University	2025
○ Selected among top graduates for comprehensive academic and extracurricular performance	
Taxman Second-Class Scholarship & Merit Student Honor, Xi'an Jiaotong University	2023
○ Awarded for academic excellence and all-round performance	
Second-Class Scholarship & Merit Student Honor, Xi'an Jiaotong University	2022
○ Honored for excellent grades and university-level achievements	

Research & Competition Achievements

First Prize, National Mathematical Modeling Competition (Shaanxi Province) 2023

- Team Leader; Advisor: Prof. Long Chen, School of Computer Science, Xi'an Jiaotong University

Second Prize, National Mathematical Modeling Competition (Shaanxi Province) 2022

- Team Leader; Advisor: Prof. Huanqin Li, School of Mathematics and Statistics, Xi'an Jiaotong University

Conference Publication: IECON 2023 (IEEE Industrial Electronics Society) 2023

- *Distributed Secondary Control for DC Microgrid with Virtual Voltage Compensation*
DOI: 10.1109/IECON51785.2023.10311983

Utility Patent Granted (National Level, China) 2023

- *A Novel Composting Instrument*; Patent No.: ZL 2023 2 3087822.1

First Prize, National English Translation Contest for College Students 2022

- Awarded for outstanding English–Chinese translation in national-level competition

Student Leadership

Head of Sports Division, Qian Xuesen College Student Union Jul 2022 – Jul 2023

- Led student sports activities and coordinated intra- and inter-college athletic events

Standing Committee Member, Qian Xuesen College Student Union Jul 2023 – Jul 2024

- Participated in strategic planning and management of the college student union

Student Delegate, 9th National Honors Education Conference May 2024

- Represented Xi'an Jiaotong University at a national-level honors talent development forum and delivered a featured presentation